R112: Full18 Locked Results

Single-seed descriptive evidence. Historical models are not newly matched controls.

Model	Split	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
R112/A_collapsed	VAL	71.060494	11.417398	11.427978	11.371692	-0.187797	0.399854	0.410621
R112/B_interfaces	VAL	71.024691	11.417900	11.427978	11.371692	-0.183590	0.404089	0.415310
R112/A_collapsed	TEST	71.283951	11.374503	11.385299	11.329932	-0.190906	0.392293	0.400308
R112/B_interfaces	TEST	71.333333	11.374755	11.385299	11.329932	-0.190510	0.392669	0.401151
R94B	TEST	71.577778	11.372234	11.385299	11.329932	-0.211169	0.371939	0.379115
R89B	TEST	71.419753	11.373470	11.385299	11.329932	-0.201446	0.381702	0.389646

R112: Corrections, Harms And Cost Gains

W=cost win rate; G=mean relative cost improvement. dRegret=current-baseline; lower is better.

Comparison	Split	Corrected	Harmed	Wrong cheaper	Wrong dearer	dTop1 pp	dRegret pp	W %	G %
B_interfaces vs A_collapsed	VAL	1252	1281	563	571	-0.035802	0.004689	2.2407	-0.006481
B_interfaces vs A_collapsed	TEST	1262	1222	575	577	0.049383	0.000843	2.2679	-0.003263
A_collapsed vs R94B	TEST	1669	1907	780	831	-0.293827	0.021193	3.0235	-0.024628
B_interfaces vs R94B	TEST	1701	1899	744	791	-0.244444	0.022036	3.0185	-0.025221
A_collapsed vs R89B	TEST	1403	1513	654	688	-0.135802	0.010662	2.5383	-0.012217
B_interfaces vs R89B	TEST	1412	1482	607	643	-0.086420	0.011505	2.4914	-0.012565

A_collapsed: VAL Per-Problem Metrics

ALL equal-problem; ratios of means differ from actual per-instance regret

Problem	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
TSP	45.933333	8.027187	8.045354	7.990533	-0.225804	0.458718	0.446902
CVRP	47.133333	18.163369	18.269377	18.064087	-0.580252	0.549608	0.731629
ATSP	77.177778	1.638411	1.676905	1.632471	-2.295516	0.363894	0.361027
OVRP	41.088889	7.462179	7.469579	7.415777	-0.099070	0.625726	0.639539
VRPB	80.822222	9.356876	9.354876	9.318750	0.021374	0.409133	0.417665
VRPL	37.577778	12.432051	12.437959	12.383444	-0.047504	0.392515	0.399350
VRPTW	80.177778	18.817987	18.819431	18.766139	-0.007676	0.276283	0.291144
OVRPTW	83.755556	10.143410	10.143730	10.116859	-0.003155	0.262439	0.260356
OVRPB	88.533333	6.357313	6.357254	6.341078	0.000920	0.256027	0.264997
OVRPL	41.200000	7.438125	7.445709	7.390416	-0.101868	0.645550	0.661734
VRPBL	79.000000	9.371116	9.371523	9.328366	-0.004336	0.458280	0.465836
VRPBTW	78.644444	19.947534	19.950430	19.851084	-0.014516	0.485871	0.463121
VRPLTW	78.933333	18.764927	18.765871	18.707843	-0.005030	0.305138	0.318795
OVRPBL	89.266667	6.369687	6.370093	6.352824	-0.006380	0.265439	0.275572
OVRPBTW	83.511111	10.660699	10.661098	10.623245	-0.003744	0.352569	0.326675
OVRPLTW	83.511111	10.157368	10.157423	10.129818	-0.000537	0.271972	0.276520
VRPBLTW	79.533333	19.794218	19.797044	19.706420	-0.014276	0.445528	0.437996
OVRPBLTW	83.288889	10.610700	10.609955	10.571302	0.007016	0.372686	0.352326
ALL	71.060494	11.417398	11.427978	11.371692	-0.187797	0.399854	0.410621

A_collapsed: TEST Per-Problem Metrics

ALL equal-problem; ratios of means differ from actual per-instance regret

Problem	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
TSP	46.955556	7.972859	7.989621	7.937593	-0.209801	0.444283	0.434692
CVRP	48.066667	18.012468	18.124983	17.918509	-0.620776	0.524369	0.689182
ATSP	77.022222	1.635269	1.674207	1.629441	-2.325774	0.357640	0.356938
OVRP	40.644444	7.445991	7.455750	7.400487	-0.130886	0.614876	0.620652
VRPB	80.244444	9.355491	9.351810	9.315298	0.039365	0.431475	0.431876
VRPL	37.844444	12.349429	12.356138	12.300958	-0.054298	0.394039	0.400380
VRPTW	81.088889	18.688367	18.690545	18.637408	-0.011655	0.273422	0.284534
OVRPTW	83.888889	10.093007	10.093063	10.068135	-0.000557	0.247032	0.238294
OVRPB	88.111111	6.371938	6.370826	6.352225	0.017467	0.310331	0.322091
OVRPL	41.688889	7.464033	7.472542	7.418033	-0.113866	0.620108	0.631026
VRPBL	79.288889	9.396092	9.396379	9.355967	-0.003056	0.428869	0.427357
VRPBTW	78.777778	19.664989	19.665725	19.577666	-0.003744	0.446034	0.440952
VRPLTW	79.244444	18.802867	18.803136	18.745143	-0.001431	0.307942	0.323378
OVRPBL	89.200000	6.375536	6.375741	6.359310	-0.003221	0.255147	0.267359
OVRPBTW	84.955556	10.658937	10.659591	10.624748	-0.006142	0.321780	0.294345
OVRPLTW	82.888889	10.085998	10.086112	10.059001	-0.001131	0.268385	0.269863
VRPBLTW	79.688889	19.705507	19.707068	19.613284	-0.007922	0.470206	0.449271
OVRPBLTW	83.511111	10.662269	10.662149	10.625574	0.001125	0.345346	0.323359
ALL	71.283951	11.374503	11.385299	11.329932	-0.190906	0.392293	0.400308

R112 | native/FP64 winners | TRAIN SBS | equal-task ALL | same locked weights; no TEST/LIB reselection

B_interfaces: VAL Per-Problem Metrics

ALL equal-problem; ratios of means differ from actual per-instance regret

Problem	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
TSP	45.800000	8.027840	8.045354	7.990533	-0.217683	0.466895	0.455467
CVRP	46.488889	18.164731	18.269377	18.064087	-0.572797	0.557148	0.740143
ATSP	77.222222	1.638522	1.676905	1.632471	-2.288885	0.370706	0.369734
OVRP	41.488889	7.461675	7.469579	7.415777	-0.105813	0.618934	0.634307
VRPB	80.911111	9.357802	9.354876	9.318750	0.031279	0.419075	0.428060
VRPL	37.533333	12.432139	12.437959	12.383444	-0.046795	0.393227	0.399406
VRPTW	80.022222	18.821240	18.819431	18.766139	0.009612	0.293620	0.313584
OVRPTW	83.733333	10.143695	10.143730	10.116859	-0.000346	0.265255	0.263193
OVRPB	88.511111	6.357154	6.357254	6.341078	-0.001569	0.253532	0.262606
OVRPL	41.422222	7.437979	7.445709	7.390416	-0.103821	0.643583	0.662451
VRPBL	78.777778	9.373808	9.371523	9.328366	0.024389	0.487138	0.491119
VRPBTW	78.644444	19.947154	19.950430	19.851084	-0.016421	0.483956	0.461343
VRPLTW	78.933333	18.764904	18.765871	18.707843	-0.005151	0.305015	0.318919
OVRPBL	89.244444	6.369656	6.370093	6.352824	-0.006862	0.264955	0.279240
OVRPBTW	83.533333	10.660566	10.661098	10.623245	-0.004993	0.351315	0.324687
OVRPLTW	83.422222	10.157395	10.157423	10.129818	-0.000273	0.272236	0.274025
VRPBLTW	79.533333	19.794411	19.797044	19.706420	-0.013303	0.446506	0.438587
OVRPBLTW	83.222222	10.611527	10.609955	10.571302	0.014814	0.380513	0.358717
ALL	71.024691	11.417900	11.427978	11.371692	-0.183590	0.404089	0.415310

R112 | native/FP64 winners | TRAIN SBS | equal-task ALL | same locked weights; no TEST/LIB reselection

B_interfaces: TEST Per-Problem Metrics

ALL equal-problem; ratios of means differ from actual per-instance regret

Problem	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
TSP	46.355556	7.973580	7.989621	7.937593	-0.200770	0.453373	0.445443
CVRP	47.733333	18.011106	18.124983	17.918509	-0.628291	0.516767	0.682839
ATSP	77.444444	1.635079	1.674207	1.629441	-2.337135	0.345966	0.346556
OVRP	40.777778	7.446297	7.455750	7.400487	-0.126791	0.619001	0.628474
VRPB	80.244444	9.354285	9.351810	9.315298	0.026462	0.418522	0.417293
VRPL	38.000000	12.349839	12.356138	12.300958	-0.050978	0.397373	0.404394
VRPTW	81.044444	18.689823	18.690545	18.637408	-0.003865	0.281234	0.292923
OVRPTW	83.888889	10.094042	10.093063	10.068135	0.009706	0.257320	0.253757
OVRPB	88.111111	6.371341	6.370826	6.352225	0.008095	0.300933	0.309105
OVRPL	42.733333	7.462407	7.472542	7.418033	-0.135629	0.598184	0.609175
VRPBL	79.200000	9.396509	9.396379	9.355967	0.001383	0.433328	0.430664
VRPBTW	78.755556	19.668384	19.665725	19.577666	0.013521	0.463377	0.455850
VRPLTW	79.244444	18.804983	18.803136	18.745143	0.009821	0.319229	0.335214
OVRPBL	89.266667	6.375710	6.375741	6.359310	-0.000495	0.257879	0.269519
OVRPBTW	84.888889	10.659126	10.659591	10.624748	-0.004362	0.323565	0.295774
OVRPLTW	82.911111	10.085941	10.086112	10.059001	-0.001693	0.267822	0.267678
VRPBLTW	79.933333	19.704431	19.707068	19.613284	-0.013380	0.464722	0.447562
OVRPBLTW	83.466667	10.662705	10.662149	10.625574	0.005213	0.349448	0.328506
ALL	71.333333	11.374755	11.385299	11.329932	-0.190510	0.392669	0.401151

R112 | native/FP64 winners | TRAIN SBS | equal-task ALL | same locked weights; no TEST/LIB reselection

B_interfaces vs A_collapsed: TEST Paired Changes

Both-wrong contributions are normalized by all problem rows; see comparison.csv for full precision.

Problem	Corrected	Harmed	Wrong cheaper	Wrong dearer	dTop1 pp	dRegret pp	W %	G %
TSP	92	119	58	81	-0.600000	0.010751	3.3333	-0.011942
CVRP	186	201	149	139	-0.333333	-0.006343	7.4444	0.001955
ATSP	62	43	8	4	0.422222	-0.010382	1.5556	0.009362
OVRP	261	255	119	113	0.133333	0.007822	8.4444	-0.009592
VRPB	34	34	4	5	0.000000	-0.014584	0.8444	-0.004455
VRPL	219	212	125	139	0.155556	0.004015	7.6444	-0.004568
VRPTW	12	14	1	2	-0.044444	0.008389	0.2889	-0.008570
OVRPTW	8	8	0	0	0.000000	0.015464	0.1778	-0.015831
OVRPB	14	14	2	3	0.000000	-0.012987	0.3556	0.006531
OVRPL	264	217	104	85	1.044444	-0.021851	8.1778	0.020060
VRPBL	24	28	3	5	-0.088889	0.003306	0.6000	-0.006421
VRPBTW	21	22	0	0	-0.022222	0.014898	0.4667	-0.015466
VRPLTW	15	15	0	1	0.000000	0.011836	0.3333	-0.012254
OVRPBL	8	5	2	0	0.066667	0.002159	0.2222	-0.002501
OVRPBTW	2	5	0	0	-0.066667	0.001429	0.0444	-0.001613
OVRPLTW	11	10	0	0	0.022222	-0.002185	0.2444	0.001237
VRPBLTW	21	10	0	0	0.244444	-0.001709	0.4667	0.001062
OVRPBLTW	8	10	0	0	-0.044444	0.005147	0.1778	-0.005736
ALL	1262	1222	575	577	0.049383	0.000843	2.2679	-0.003263

A_collapsed vs R94B: TEST Paired Changes

Both-wrong contributions are normalized by all problem rows; see comparison.csv for full precision.

Problem	Corrected	Harmed	Wrong cheaper	Wrong dearer	dTop1 pp	dRegret pp	W %	G %
TSP	135	138	82	81	-0.066667	0.026806	4.8222	-0.027482
CVRP	259	277	186	204	-0.400000	0.020639	9.8889	-0.025513
ATSP	37	44	8	2	-0.155556	0.002216	1.0000	-0.002744
OVRP	341	333	162	176	0.177778	0.000341	11.1778	-0.003132
VRPB	46	83	8	4	-0.822222	0.077494	1.2000	-0.081798
VRPL	280	272	175	177	0.177778	0.000205	10.1111	-0.000879
VRPTW	27	39	0	2	-0.266667	0.026795	0.6000	-0.027231
OVRPTW	17	26	0	0	-0.200000	-0.009897	0.3778	-0.003038
OVRPB	23	33	2	5	-0.222222	0.064107	0.5556	-0.066160
OVRPL	330	335	143	166	-0.111111	0.010884	10.5111	-0.012817
VRPBL	44	95	7	6	-1.133333	0.059254	1.1333	-0.062093
VRPBTW	36	51	0	0	-0.333333	0.015515	0.8000	-0.016802
VRPLTW	23	34	2	0	-0.244444	0.011367	0.5556	-0.015858
OVRPBL	15	29	4	7	-0.311111	0.024929	0.4222	-0.026837
OVRPBTW	9	15	0	0	-0.133333	0.009872	0.2000	-0.012336
OVRPLTW	14	31	1	1	-0.377778	-0.006963	0.3333	-0.008422
VRPBLTW	25	51	0	0	-0.577778	0.030203	0.5556	-0.032230
OVRPBLTW	8	21	0	0	-0.288889	0.017704	0.1778	-0.017933
ALL	1669	1907	780	831	-0.293827	0.021193	3.0235	-0.024628

B_interfaces vs R94B: TEST Paired Changes

Both-wrong contributions are normalized by all problem rows; see comparison.csv for full precision.

Problem	Corrected	Harmed	Wrong cheaper	Wrong dearer	dTop1 pp	dRegret pp	W %	G %
TSP	127	157	77	94	-0.666667	0.037558	4.5333	-0.038172
CVRP	276	309	209	216	-0.733333	0.014296	10.7778	-0.019644
ATSP	76	64	15	5	0.266667	-0.008165	2.0222	0.006855
OVRP	336	322	141	155	0.311111	0.008163	10.6000	-0.010417
VRPB	46	83	6	1	-0.822222	0.062910	1.1556	-0.067174
VRPL	268	253	142	161	0.333333	0.004219	9.1111	-0.004839
VRPTW	28	42	1	3	-0.311111	0.035184	0.6444	-0.035496
OVRPTW	15	24	0	0	-0.200000	0.005567	0.3333	-0.018343
OVRPB	21	31	2	6	-0.222222	0.051120	0.5111	-0.052897
OVRPL	336	294	138	134	0.933333	-0.010966	10.5333	0.008829
VRPBL	40	95	7	9	-1.222222	0.062560	1.0444	-0.065256
VRPBTW	30	46	0	0	-0.355556	0.030413	0.6667	-0.031266
VRPLTW	25	36	1	0	-0.244444	0.023203	0.5778	-0.023716
OVRPBL	15	26	4	6	-0.244444	0.027088	0.4222	-0.029113
OVRPBTW	9	18	0	0	-0.200000	0.011301	0.2000	-0.013901
OVRPLTW	16	32	1	1	-0.355556	-0.009148	0.3778	-0.006426
VRPBLTW	28	43	0	0	-0.333333	0.028494	0.6222	-0.029896
OVRPBLTW	9	24	0	0	-0.333333	0.022851	0.2000	-0.023104
ALL	1701	1899	744	791	-0.244444	0.022036	3.0185	-0.025221

A_collapsed vs R89B: TEST Paired Changes

Both-wrong contributions are normalized by all problem rows; see comparison.csv for full precision.

Problem	Corrected	Harmed	Wrong cheaper	Wrong dearer	dTop1 pp	dRegret pp	W %	G %
TSP	133	140	73	75	-0.155556	0.023207	4.5778	-0.024018
CVRP	208	234	159	175	-0.577778	0.022056	8.1556	-0.028680
ATSP	66	61	1	5	0.111111	0.000842	1.4889	-0.001741
OVRP	309	302	136	146	0.155556	-0.012258	9.8889	0.008557
VRPB	26	46	3	3	-0.444444	0.067192	0.6444	-0.069278
VRPL	237	258	159	148	-0.466667	0.005881	8.8000	-0.006611
VRPTW	10	14	0	0	-0.088889	-0.001198	0.2222	0.000668
OVRPTW	2	4	0	0	-0.044444	0.003503	0.0444	-0.003542
OVRPB	15	20	1	1	-0.111111	0.039213	0.3556	-0.040883
OVRPL	294	291	120	129	0.066667	0.013736	9.1778	-0.015353
VRPBL	30	38	1	3	-0.177778	0.014201	0.6889	-0.014998
VRPBTW	25	26	0	0	-0.022222	-0.015096	0.5556	0.009387
VRPLTW	8	16	1	0	-0.177778	0.007089	0.2000	-0.007282
OVRPBL	10	13	0	3	-0.066667	-0.003149	0.2222	0.001983
OVRPBTW	3	2	0	0	0.022222	-0.000436	0.0667	0.000170
OVRPLTW	6	11	0	0	-0.111111	0.008123	0.1333	-0.008502
VRPBLTW	18	26	0	0	-0.177778	0.012355	0.4000	-0.013049
OVRPBLTW	3	11	0	0	-0.177778	0.006659	0.0667	-0.006738
ALL	1403	1513	654	688	-0.135802	0.010662	2.5383	-0.012217

B_interfaces vs R89B: TEST Paired Changes

Both-wrong contributions are normalized by all problem rows; see comparison.csv for full precision.

Problem	Corrected	Harmed	Wrong cheaper	Wrong dearer	dTop1 pp	dRegret pp	W %	G %
TSP	113	147	62	82	-0.755556	0.033959	3.8889	-0.034650
CVRP	209	250	160	158	-0.911111	0.015713	8.2000	-0.019618
ATSP	69	45	3	3	0.533333	-0.009539	1.6000	0.008518
OVRP	305	292	118	132	0.288889	-0.004436	9.4000	0.001635
VRPB	18	38	1	1	-0.444444	0.052609	0.4222	-0.053075
VRPL	238	252	134	143	-0.311111	0.009896	8.2667	-0.010560
VRPTW	13	19	1	1	-0.133333	0.007192	0.3111	-0.007544
OVRPTW	5	7	0	0	-0.044444	0.018967	0.1111	-0.019006
OVRPB	10	15	1	2	-0.111111	0.026226	0.2444	-0.027336
OVRPL	326	276	123	111	1.111111	-0.008114	9.9556	0.006205
VRPBL	26	38	2	6	-0.266667	0.017507	0.6222	-0.019674
VRPBTW	19	21	0	0	-0.044444	-0.000198	0.4222	-0.001033
VRPLTW	14	22	1	1	-0.177778	0.018925	0.3333	-0.019286
OVRPBL	8	8	1	3	0.000000	-0.000989	0.2000	0.000039
OVRPBTW	3	5	0	0	-0.044444	0.000993	0.0667	-0.001118
OVRPLTW	10	14	0	0	-0.088889	0.005939	0.2222	-0.006317
VRPBLTW	22	19	0	0	0.066667	0.010645	0.4889	-0.011260
OVRPBLTW	4	14	0	0	-0.222222	0.011806	0.0889	-0.012082
ALL	1412	1482	607	643	-0.086420	0.011505	2.4914	-0.012565

R112: Same-Checkpoint Native LIB

TEST participated in earlier R&D; not a pristine blind test; no finish-stage reselection

Model	Split	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
R112/A_collapsed	TSPLIB	32.653061	8.196547	8.240435	8.142167	-0.532601	0.667876	0.686449
R112/A_collapsed	CVRPLIB	64.000000	67.242449	67.245364	66.945093	-0.004335	0.444179	0.538517
R112/B_interfaces	TSPLIB	26.530612	8.199875	8.240435	8.142167	-0.492212	0.708753	0.721840
R112/B_interfaces	CVRPLIB	63.000000	67.256356	67.245364	66.945093	0.016345	0.464952	0.530228

R112: Provenance And Scientific Boundary

same seeded task/shuffle protocol; actual per-update sample IDs not independently archived

Item	Evidence
Logical batch	512
Initial trainable SHA	3a0b459f6e5e99c28ce5941054c610ce9fa8dcc09ed725bef327898f322e4d00
A_collapsed checkpoint	4bf692f36fde2f3f7ce461ccc96a01b2bd254e606c9ea40c7f5e3aea9c44ae3e
B_interfaces checkpoint	55ccef0ca150c28311cf9340d740a4f0cac80f3a921b1d8f3cbe07fd9017ecb5
Data version	selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018
R94B	complete:
R89B	complete:
Seed	2 only; no multi-seed significance claim
Mechanism	identical fresh tensors/shapes; collapse versus retained interface conditions
Source of metrics	immutable saved predictions; no new forward passes or calibration